

Multiple Choice Questions (MCQs)

DIRECTIONS : This section contains multiple choice questions. Each question has four choices (a), (b), (c) and (d) out of which only one is correct.

- On cutting a solenoid in half, the field lines remain ...A..., emerging from one face of the solenoid and entering into the other face. Here, A refers to
(a) irregular (b) discontinuous
(c) continuous (d) alternate
- An iron rod of length L and magnetic moment M is bent in the form of a semicircle. Now its magnetic moment will be
(a) M (b) $2M/\pi$ (c) M/π (d) M/π
- Which of the following statement(s) is/are true/false about magnetism?
I. The Earth behaves as a magnet with the magnetic field pointing approximately from the geographic South to the North.
II. When a bar magnet is freely suspended, it points in the North–South direction. The tip which points to the geographic North is called the North –pole and the tip which points to geographic South is called the South–pole of magnet.
III. There is a repulsive force when North–poles (or South–poles) of two magnets are brought close together. Conversely, there is an attractive force between the North–pole of one magnet and the South–pole of other.
IV. We can isolate the North or South–pole of a magnet.
(a) T, T, F, F (b) T, T, F, T
(c) F, F, T, T (d) T, T, T, F
- The horizontal component of the earth's magnetic field is 3.6×10^{-5} tesla where the dip angle is 60° . The magnitude of the earth's magnetic field is
(a) 2.8×10^{-4} tesla (b) 2.1×10^{-4} tesla
(c) 7.2×10^{-5} tesla (d) 3.6×10^{-5} tesla
- At a certain place, horizontal component is $\sqrt{3}$ times the vertical component. The angle of dip at this place is
(a) 0 (b) $\pi/3$ (c) $\pi/6$ (d) $\pi/8$
- Let V and H be the vertical and horizontal components of earth's magnetic field at any point on earth. Near the north pole
(a) $V \gg H$ (b) $V \ll H$ (c) $V = H$ (d) $V = H = 0$
- If μ_0 is absolute permeability of vacuum and μ_r is relative magnetic permeability of another medium, then permeability μ of the medium is
(a) $\mu_0 \mu_r$ (b) μ_0/μ_r (c) μ_r/μ_0 (d) $1/\mu_0 \mu_r$
- A toroid of n turns, mean radius R and cross-sectional radius a carries current I . It is placed on a horizontal table taken as xy -plane. Its magnetic moment m
(a) is non-zero and points in the z -direction by symmetry
(b) points along the axis of the toroid ($m = m\phi$)
(c) is zero, otherwise there would be a field falling as $\frac{1}{r^3}$ at large distances outside the toroid
(d) is pointing radially outwards
- The magnetic field of the earth can be modelled by that of a point dipole placed at the centre of the earth. The dipole axis makes an angle of 11.3° with the axis of the earth. At Mumbai, declination is nearly zero. Then,
(a) the declination varies between 11.3° W to 11.3° E
(b) the least declination is 0°
(c) the plane defined by dipole axis and the earth axis passes through Greenwich
(d) declination averaged over the earth must be always negative
- A bar magnet is cut into two equal halves by a plane parallel to the magnetic axis. Of the following physical quantities the one which remains unchanged is
(a) pole strength
(b) magnetic moment
(c) intensity of magnetisation
(d) None of these
- One can define ...A... of a place as the vertical plane which passes through the imaginary line joining the magnetic North and the South–poles. Here, A refers to
(a) geographic meridian (b) magnetic meridian
(c) magnetic declination (d) magnetic inclination
- The magnetic compass is not useful for navigation near the magnetic poles, since
(a) $R=0$ (b) $V=0$ (c) $H=0$ (d) $\theta=0^\circ$
- The magnetism of magnet is due to
(a) spin motion of electron
(b) earth
(c) pressure of big magnet inside the earth
(d) cosmic rays

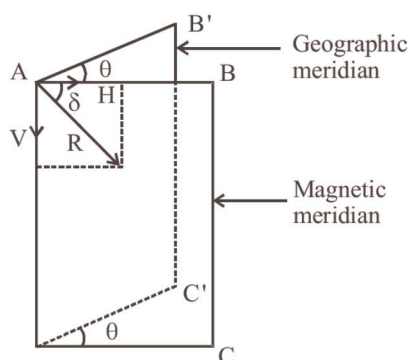
14. The primary origin (s) of magnetism lies in
 - (a) atomic currents (b) polar nature of molecules
 - (c) extrinsic spin of electron (d) None of these
15. Which of these is correct in regard to a magnet?
 - (a) Geometric length = 0.8 times the magnetic length
 - (b) Magnetic length = 0.8 times the geometric length
 - (c) Magnetic length = Geometric length
 - (d) Geometric length = 10/9 of magnetic length
16. Magnetic dipole moment is a vector quantity directed from
 - (a) south pole to north pole (b) north pole to south pole
 - (c) east to west (d) west to east
17. A bar magnet of magnetic moment M and length L is cut into two equal parts each of length $L/2$. The magnetic moment of each part will be
 - (a) M (b) $M/4$ (c) $\sqrt{2} M$ (d) $M/2$
18. When a current in a circular loop is equivalently replaced by a magnetic dipole
 - (a) the pole strength m of each pole is fixed
 - (b) the distance d between the poles is fixed
 - (c) the product md is fixed
 - (d) None of these
19. The magnetic moment of atomic neon is equal to
 - (a) zero (b) $\frac{1}{2} \mu_B$ (c) μ_B (d) $\frac{3}{2} \mu_B$
20. Current i is flowing in a coil of area A and number of turns N , then magnetic moment of the coil, M is
 - (a) NiA (b) $\frac{Ni}{A}$ (c) $\frac{Ni}{\sqrt{A}}$ (d) N^2Ai
21. Magnetic field intensity is defined as
 - (a) Magnetic moment per unit volume
 - (b) Magnetic induction force acting on a unit magnetic pole
 - (c) Number of lines of force crossing per unit area
 - (d) Number of lines of force crossing per unit volume
22. The magnetic moment of a bar magnet is thus ... A ... to the magnetic moment of an equivalent solenoid that produces the same magnetic field. Here, A refers to
 - (a) unequal (b) different (c) equal (d) same
23. The magnetic lines of force inside a bar magnet
 - (a) are from north-pole to south-pole of the magnet
 - (b) do not exist
 - (c) depend upon the area of cross-section of the bar magnet
 - (d) are from south-pole to north-pole of the Magnet
24. The strength of the earth's magnetic field is
 - (a) constant everywhere
 - (b) zero everywhere
 - (c) having very high value
 - (d) vary from place to place on the earth's surface
25. The correct relation is
 - (a) $B = \frac{B_V}{B_H}$ (b) $B = B_V \times B_H$
 - (c) $|B| = \sqrt{B_H^2 + B_V^2}$ (d) $B = B_H + B_V$
26. The line on the earth surface joining the point where the field is horizontal, is called
 - (a) magnetic equator (b) magnetic line
 - (c) magnetic axis (d) magnetic inertia
27. Horizontal component of earth's field at a height of 1 m from the surface of earth is H . Its value at a height of 10 m from surface of earth is
 - (a) $H/10$ (b) $H/9$ (c) $H/100$ (d) H
28. At the magnetic north pole of the earth, the value of the horizontal component of earth's magnetic field and angle of dip are respectively
 - (a) zero, maximum (b) maximum, minimum
 - (c) maximum, maximum (d) minimum, minimum
29. Which of the following is responsible for the earth's magnetic field?
 - (a) Convective currents in earth's core.
 - (b) Divergent current in earth's core.
 - (c) Rotational motion of earth.
 - (d) Translational motion of earth.
30. Demagnetisation of magnets can be done by
 - (a) rough handling
 - (b) heating
 - (c) magnetising in the opposite direction
 - (d) All the above
31. A compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It will
 - (a) stay in north-south direction only
 - (b) stay in east-west direction only
 - (c) become rigid showing no movement
 - (d) stay in any position
32. The time period of oscillation of a freely suspended bar magnet with usual notations is given by
 - (a) $T = 2\pi \sqrt{\frac{I}{MB_H}}$ (b) $T = 2\pi \sqrt{\frac{MB_H}{I}}$
 - (c) $T = \sqrt{\frac{I}{MB_H}}$ (d) $T = 2\pi \sqrt{\frac{B_H}{MI}}$
33. To measure the magnetic moment of a bar magnet, one may use
 - (a) a deflection galvanometer if the earth's horizontal field is known
 - (b) an oscillation magnetometer if the earth's horizontal field is known
 - (c) both deflection and oscillation magnetometer if the earth's horizontal field is not known.
 - (d) all of the above
34. For protecting a sensitive equipment from the external electric arc, it should be
 - (a) wrapped with insulation around it when a current is passing through it
 - (b) placed inside an iron core
 - (c) surrounded with fine copper sheet
 - (d) placed inside an aluminium can
35. The magnetic needle has magnetic moment $8.7 \times 10^{-2} \text{ Am}^2$ and moment of inertia $11.5 \times 10^{-6} \text{ kgm}^2$. It performs 10 complete oscillations in 6.70 s, what is the magnitude of the magnetic field?
 - (a) 0.012 T (b) 0.120 T (c) 1.200 T (d) 2.10 T

Case/Passage Based Questions

DIRECTIONS : Study the given Case/Passage and answer the following questions.

Case/Passage

Magnetic elements of earth at a place are the quantities which can completely describe in magnitude and direction, the magnetic field of the earth. There are three elements of the magnetic field of the earth.



Horizontal component of total intensity, $B_H = R \cos \delta$;

Vertical component of total intensity,

$$B_V = R \sin \delta \Rightarrow B_H^2 + B_V^2 = R^2$$

$$\therefore R = \sqrt{B_H^2 + B_V^2} \text{ and } \tan \delta = \frac{B_V}{B_H}$$

36. The strength of the earth's magnetic field is
 (a) constant everywhere
 (b) zero everywhere
 (c) having very high value
 (d) vary from place to place on the earth's surface
37. The lines of force due to earth's horizontal magnetic field are
 (a) parallel and straight (b) concentric circles
 (c) elliptical (d) curved lines
38. The earth's magnetic field always has a vertical component except at the
 (a) magnetic equator (b) magnetic poles
 (c) geographic north pole (d) latitude 45°
39. At magnetic poles, the angle of dip is
 (a) 45° (b) 30° (c) zero (d) 90°
40. At the magnetic north pole of the earth, the value of the horizontal component of earth's magnetic field and angle of dip are respectively
 (a) zero, maximum (b) maximum, minimum
 (c) maximum, maximum (d) minimum, minimum

Assertion & Reason

DIRECTIONS : Each of these questions contains an assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and the Reason is the **correct explanation** of the Assertion.
 (b) If both **Assertion** and **Reason** are correct but Reason is **not the correct explanation** of the Assertion.
 (c) If the **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If the **Assertion** is **incorrect** but the **Reason** is **correct**.

41. **Assertion :** Magnetic moment of an atom is due to both the orbital motion and spin motion of every electron.
Reason : A charged particle produces magnetic field.
42. **Assertion :** The poles of magnet can not be separated by breaking into two pieces.
Reason : The magnetic moment will be reduced to half when a magnet is broken into two equal pieces.
43. **Assertion :** Soft iron is used as transformer core.
Reason : Soft iron has narrow hysteresis loop.

44. **Assertion :** The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil.

Reason : Soft iron has high magnetic permeability and cannot be easily magnetized or demagnetized.

Match the Following

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (1, 2, 3, 4) in column-II.

45. Match the column-I and column II.

Column I	Column II
(A) Horizontal component	(1) $B_E \sin \phi$
(B) Vertical component	(2) $\frac{B_V}{B_H}$
(C) $\tan \phi$	(3) $B_E \cos \phi$
(D) Tangent law	(4) $B = B_H \tan \theta$
(a) $A \rightarrow (3); B \rightarrow (2); C \rightarrow (1); D \rightarrow (4)$	
(b) $A \rightarrow (3); B \rightarrow (1); C \rightarrow (2); D \rightarrow (4)$	
(c) $A \rightarrow (2); B \rightarrow (3); C \rightarrow (1); D \rightarrow (4)$	
(d) $A \rightarrow (1); B \rightarrow (3); C \rightarrow (2); D \rightarrow (4)$	

Fill in the Blanks

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

46. If horizontal and vertical components of earth's magnetic field are equal, then angle of dip is _____.
47. At a certain place, the angle of dip is 30° and the horizontal component of earth's magnetic field is 0.50 oersted. The earth's total magnetic field (in oersted) is _____.
48. The period of oscillation of a magnet in a vibration magnetometer is 2 sec. The period of oscillation of a magnet whose magnetic moment is four times that of the first magnet is _____.
49. When 2 ampere current is passed through a tangent galvanometer, it gives a deflection of 30° . For 60° deflection, the current must be _____.
50. The magnetic field and angle of dip at a place on the earth are 0.3 G and 30° respectively. The value of vertical component of the earth's magnetic field at the place is _____. [CBSE 2020]

True / False

DIRECTIONS : Read the following statements and write your answer as true or false.

51. The area of hysteresis loop for steel is large as compared to soft iron.
52. Steel dissipate greater energy than soft iron.
53. If a compass needle be kept at magnetic north pole of the earth, the compass needle may stay in any direction
54. Dip needle will stay vertical at the north pole of earth.

ANSWER KEY & SOLUTIONS

1. (c) The field lines remain continuous, emerging from one face of the solenoid and entering into the other face.
2. (b) On bending a rod its pole strength remain unchanged where as its magnetic moment changes new magnetic moment $M'_{m(2R)} = \left(\frac{2L}{\pi}\right) = \frac{2M}{\pi}$.

3. (d) (i) The Earth behaves as a magnet with the magnetic field pointing appoximately from the geographic South to the North.

(ii) When a bar magnet is freely suspended, it points in the North-South direction. The tip which points to the geographic North is called the North pole and the tip which points to the geographic South is called the South-pole of the magnet.

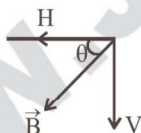
(iii) There is a repulsive force when North poles (or South-poles) of two magnets are brought close together. Conversely there is an attractive force between the North-pole of one magnet and the South-pole of the other.

(iv) We cannot isolate the North or South-pole of a magnet. If a bar magnet is broken into two halves, we get two similar bar magnets with somewhat weaker properties. Unlike electric charges, isolated magnetic North and South poles known as magnetic monopoles do not exist.

4. (c) Horizontal component of earth's field,
 $H = B \cos \theta$, since, $\theta = 60^\circ$

$$3.6 \times 10^{-5} = B \times \frac{1}{2}$$

$$\Rightarrow B = 7.2 \times 10^{-5} \text{ Tcsla}$$



5. (c) $\tan \delta = \frac{V}{H} = \frac{V}{\sqrt{3}V} = \frac{1}{\sqrt{3}}$

$$\therefore \delta = 30^\circ = \pi/6 \text{ radian}$$

6. (a)

7. (a) Relative magnetic permeability

$$\mu_r = \frac{\mu}{\mu_0} \Rightarrow \mu = \mu_0 \times \mu_r$$

8. (c) Toroid is a hollow circular ring on which a large number of turns of a wire are closely wound. Thus, in this case magnetic field is only confined inside the body of toroid. So no magnetic field outside the toroid and magnetic field only inside the toroid.

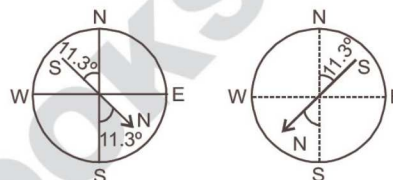
In case of toroid, the magnetic field is in the form of concentric magnetic lines of force and there is no magnetic field outside the body of toroid. This is because the loop encloses no current. Thus, the magnetic moment of toroid is zero.

In other case, if we take r as a large distance outside the toroid, then $m \propto \frac{1}{r^3}$. Which is not possible.

9. (a) Magnetic declination is an angle between angle of magnetic meridian and the geographic meridian.

As the earth's magnetism, the magnetic field lines of the earth resemble that of a hypothetical magnetic dipole located at the centre of the earth.

The axis of the dipole does not coincide with the axis of rotation of the earth but is presently tilted by 11.3° (approx) with respect to geographical axis earth. This results into two situations as given in the figure.



So, the declination varies between 11.3° W to 11.3° E .

10. (c)

11. (b) Magnetic meridian of a place is defined as the vertical plane which passes through the imaginary line joining the magnetic North and South-poles. This plane would intersect the surface of the Earth in a longitude like circle.

12. (c)

13. (a) Spin motion of electron

14. (a)

15. (b) Magnetic length of a magnet is roughly 0.8 times the geometric length.

16. (a)

17. (d) As magnetic moment = pole strength $\times$ length and length is halved without affecting pole strength, therefore, magnetic moment becomes half.

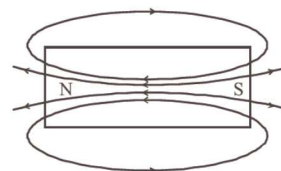
18. (c)

19. (a) Magnetic moment is cancelled and $\mu_{\text{net}} = 0$.

20. (a) 21. (b)

22. (c) The magnetic moment of a bar magnet is thus equal to the magnetic moment of an equivalent solenoid that produces the same magnetic field.

23. (d) As shown in the figure, the magnetic lines of force are directed from south to north inside a bar magnet.



24. (d) The strength of the earth's magnetic field is not constant. It varies from one place to other place on the surface of earth. Its value being of the order of 10^{-5} T .

Magnetism and Matter

25. (c) 26. (a)

27. (d) The value of H is fairly uniform.

28. (a)

29. (a) The earth's core is hot and molten. Hence, convective current in earth's core is responsible for its magnetic field.

30. (d) 31. (d) 32. (a) 33. (d)

34. (b) The iron core produces a magnetic screening for the equipment as lines of magnetic force can not enter iron enclosure.

35. (a) Magnetic moment, $M = 8.7 \times 10^{-2} \text{ Am}^2$ moment of inertia, $I = 11.5 \times 10^{-6} \text{ kg m}^2$ Time period of oscillation is

$$T = \frac{6.70}{10} = 0.6755$$

$$\text{As, } T = 2\pi \sqrt{\frac{I}{MB}}; B = \frac{4\pi^2 I^2}{MT^2}$$

$$\therefore B = \frac{4 \times (3.14)^2 \times 11.5 \times 10^{-6}}{8.7 \times 10^{-2} \times (0.67)^2} = 0.012 \text{ T}$$

36. (d) The strength of the earth's magnetic field is not constant. It varies from one place to other place on the surface of earth. Its value being of the order of 10^{-5} T .

37. (b) 38. (a) 39. (d) 40. (a)

41. (c) In an atom, electrons revolve around the nucleus and such the circular orbits of electrons may be considered as the small current loops. In addition to orbital motion, an electron has got spin motion also. So the total magnetic moment of electron is the vector sum of its magnetic charge moments due to orbital and spin motion. Particles at rest do not produce magnetic field.

42. (b) When a magnet is cut into pieces, each piece becomes new magnet. $M' = \frac{m\ell}{2} = \frac{M}{2}$.

43. (a)

44. (c) Sensitivity of galvanometer, $s = \frac{\theta}{i} \approx \frac{\tan \theta}{i} = \frac{\mu_0 N}{2RB_H}$.

If a magnetic material is placed inside coil of galvanometer, then

$$s' = \frac{\mu_r \mu_0 N}{2RB_H}$$

45. (b) $B_H = B_E \cos \phi$

$$B_V = B_E \sin \phi$$

$$\tan \phi = \frac{B_V}{B_H} \text{ and } B = B_H \tan \theta$$

where, θ = magnetic declination ϕ = Angle of inclination or dip angle46. (45°) $\tan \theta = \frac{B_V}{B_H} = 1, B_V = B_H$

$$\theta = \tan^{-1}(1) = 45^\circ$$

47. $(1/\sqrt{3}) \quad B = \frac{H}{\cos \theta} = \frac{0.50}{\cos 30^\circ} = \frac{0.50 \times 2}{\sqrt{3}} = 1/\sqrt{3}$ 48. (1s) $T = 2\pi \sqrt{\left(\frac{I}{MB_H}\right)}$

$$T' = 2\pi \sqrt{\left(\frac{I}{4MB_H}\right)} = \frac{1}{2} \left[2\pi \sqrt{\frac{I}{MB_H}} \right]$$

$$= \frac{1}{2} \times 2 = 1 \text{ second.}$$

49. (6A) As $\frac{i_2}{i_1} = \frac{\tan \theta_2}{\tan \theta_1}$

50. (0.15G) Vertical component of magnetic field

$$B_V = B \sin \delta = 0.3 \sin 30^\circ = 0.3 \times \frac{1}{2} = 0.15 \text{ G}$$

51. (True)

52. (True)

53. (True) We know that at magnetic poles the horizontal component of earth's field is zero, only vertical component exists. So compass needle may stay in any direction.

54. (True) The dip needle rotates in a vertical plane and the angle of dip at poles is 90° . Hence, the dip needle will stand vertical at the north pole of earth.